

Nikhil Madeti

+65 9121 2849 | nikhil018@e.ntu.edu.sg | Singaporean
[linkedin.com/in/nikhil-madeti](https://www.linkedin.com/in/nikhil-madeti) | github.com/Nikhil-0

EDUCATION

Nanyang Technological University, Singapore

Aug 2024 - May 2028

Bachelor of Computing (Hons), Computer Science

Coursework: Data Structures & Algorithms, Object-Oriented Design, Databases, Operating Systems, Computer Networks, Statistics

EXPERIENCE

SP Group - Software Engineer Intern

Jul 2026 - Present

- Benchmarked PostgreSQL against MongoDB Time Series across 36M records of solar generation data for a new grid-monitoring module, pairing latency results with cost projections and presenting the comparison that the team used to settle its data store choice.
- Eliminated race conditions across 3 Digital Twin mutation operations by introducing an object-level lock attribute, closing a concurrency gap reachable well beyond the cable-deletion flow the original ticket scoped.
- Cut reads per cable deletion by one full iteration (3 to 2, 33% fewer) while repairing a broken deletion flow in the Digital Twin, by shifting collection of deactivated cable IDs into the existing CSV scan.
- Removed a manual container teardown from every restart of the internal AI chatbot POC, consolidating a two-step process into a single command and unblocking 10 engineers from a recurring startup failure.

Accenture - Data Engineer Intern

May 2026 - Jul 2026

- Upgraded 14 AWS Lambda services from Python 3.9 to 3.12 and Node.js 14 to 22 during a public sector client's Commercial Cloud to Government on Commercial Cloud migration, resolving breaking changes and bringing every function onto a supported LTS version before cutover.
- Redeployed that fleet into a regulated, high-availability Government Cloud environment, each service owning one bounded capability such as location lookup and location-history retrieval.
- Caught 8 defects pre-deployment by building the test suite covering unit coverage on Lambda functions and integration flows across API Gateway, DynamoDB, S3 and GraphQL, then briefed public sector client stakeholders on the resulting architecture decisions.

PROJECTS

AI-Powered Market Analysis - Python, scikit-learn, XGBoost, spaCy, Plotly/Dash

Oct 2025

- Built an end-to-end research pipeline pulling daily OHLCV data from Yahoo Finance and 100 Finviz headlines per ticker, scoring sentiment with a VADER lexicon extended by 43 finance-specific terms and extracting company entities via spaCy NER into configurable co-occurrence networks.
- Compared logistic regression, Random Forest and XGBoost on next-day price direction under walk-forward time-series cross-validation; coefficient analysis showed headline sentiment carried near-zero weight ($\sim 1e-4$) against price and volume features (0.20-0.35), quantifying the limits of daily sentiment as a predictive signal.
- Shipped an interactive Plotly/Dash dashboard with adjustable entity-network thresholds for exploring news flow and cross-company narratives.
- GitHub: github.com/Nikhil-0/Market-Analysis

Multi-Agent Research Assistant - Google ADK, Cloud Run, Cloud Build

Oct - Nov 2025

- Built a multi-agent research assistant on Google ADK that decomposes a research question across 4 specialist agents and synthesises their outputs into a single brief, exposed through both a CLI and a web interface.
- GitHub: <https://github.com/Nikhil-0/Research-Assistant-Application>

BTO Housing Application System - Java, OOP design patterns (team of 5)

Mar - Apr 2025

- Built a role-based HDB flat application system in Java (41 classes, $\sim 4,100$ LOC) serving Applicant, Officer and Manager roles, each resolving to a distinct permission set and menu flow at runtime through a Strategy pattern keyed on user role.
- Modelled the application lifecycle as an explicit state machine (pending $\rightarrow$ successful $\rightarrow$ booked) with eligibility rules enforced in the domain layer (age and marital-status constraints on flat type) and unit inventory guarded against overbooking.
- GitHub: <https://github.com/Nikhil-0/CLI-based-BTO-system>

SKILLS

Languages: Java, Python, Go, C++, SQL, JavaScript

Backend and Data: Spring Boot, Node.js, PostgreSQL, MongoDB, MySQL, REST & GraphQL APIs

Infrastructure: AWS (Lambda, DynamoDB, S3, API Gateway), Docker, Git, CI/CD, Linux

Practices: Unit and integration testing, Agile/Scrum, code review

Certifications: Bloomberg Market Concepts

Spoken: English, Hindi, Telugu · Mandarin, Japanese (beginner)